Prithvi Paturi

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Education

Wilfrid Laurier University — Waterloo, ON
Bachelor of Science in Computer Science, Minor in Mathematics — *Sept 2022 – Apr 2027*

- **Concentration:** Big Data Systems
- **Relevant Coursework:** Software Engineering, Databases, Data Structures, Operating Systems, Computer Networks, Algorithm Design & Analysis, Windows Application Programming

Experience

Royal Bank of Canada — Toronto, ON
Incoming Software Engineer Intern — *Sept 2026 – Apr 2027*

Clio — Toronto, ON
Software Developer Intern — *Jan 2026 – Mar 2026*

- Shipped **3** features across Clio mobile applications, used by **150,000+** users, improving usability in high-traffic workflows
- Investigated high-latency API endpoints using **Datadog** tracing and logs; identified bottlenecks in service-to-service calls and database query patterns, reducing API timeout and failure incidents by **60%**
- Engineered **RESTful Ruby on Rails** endpoints integrating **AWS S3** and **AWS Lambda** for asynchronous file-processing workflows; reduced API error rates from **8%** to **1.5%** through improved validation and retry handling
- Drove the development of a **React Native** mobile platform for **Vincent AI** during a **48-hour** hackathon; concept was selected for continued development and integrated into **Clio's** product roadmap

CIBC Capital Markets — Toronto, ON
Software Developer Intern — *May 2024 – Dec 2024*

- Built **Python** and **Java** microservices on **Azure** to automate cross-team reporting pipelines, automating cross-team reporting workflows, and eliminating repetitive manual reporting tasks
- Designed and implemented automated API testing frameworks using **REST Assured**, increasing QA coverage by **30%** and improving API reliability and regression coverage across services
- Built **CI/CD** monitoring dashboards in **Azure DevOps**, reducing incident detection and response time by **40%**
- Integrated **CI/CD pipelines** across teams, reducing manual deployment effort by **60%** and improving release consistency

Fresh Water Foods Inc. — Richmond Hill, ON
Web Developer — *Jan 2024 – Apr 2024*

- Built and deployed a responsive business website using **HTML**, **CSS**, and **Bootstrap**, establishing an online presence
- Worked directly with the owner to create a user-friendly interface to ensure easy navigation and product display

Projects

AI Fitness Form Detection App | *Python, React Native, MediaPipe, Supabase, FastAPI, scikit-learn*

- Built full-stack fitness app with **React Native**, **FastAPI**, and **Supabase/PostgreSQL** for auth, tracking, and uploads
- Developed pose analysis pipeline using **MediaPipe** Pose, **OpenCV**, and **NumPy** to detect reps and identify ROM issues
- Engineered backend APIs for workout processing and exercise feedback with token-based auth and async upload handling

NBA Player Clustering & Similarity Engine | *Python, Flask, scikit-learn, Pandas, NumPy, Matplotlib*

- Built an ML-powered player similarity engine using clustering and dimensionality reduction with advanced statistics
- Engineered normalized playstyle vectors and applied **K-Means** and **cosine similarity** to quantify stylistic similarity
- Built a modular preprocessing and **inference pipeline**, preparing for **REST API** integration and cloud deployment

Personal Finance Dashboard | *Python, JavaScript, Django, Chart.js, NumPy*

- Built a full-stack personal finance tracker using **Django** backend and **Chart.js** dashboards for real-time expense analytics
- Implemented custom **Isolation Forest** and **K-Means** models to flag unusual spending patterns

Technical Skills

Languages: Python, Ruby, Java, JavaScript, TypeScript, SQL
Frameworks/Technologies: React, React Native, Node.js, Django, Flask, FastAPI, Ruby on Rails
Cloud & Backend: AWS (S3, Lambda), Azure, Supabase
Databases: MySQL, PostgreSQL, MongoDB
DevOps & CI/CD: Git, GitHub Actions, Azure DevOps, Azure Pipelines